

2. (a) What are the various types of Milling Cutters that are used in milling? 5
- (b) What is the marking system followed in case of grinding wheels? Explain the individual elements of the marking system from the standpoint of the functioning of the wheel. 4
- (c) Explain the concept of location. Also explain the various types of locators used in jigs and fixtures. 5
3. (a) Explain the TIG and MIG systems of arc-welding. Enumerate the applications of each. 7
- (b) Explain the principle underlying the resistance welding process. Enlist the names of products where in the following processes are used:
 - (i) Spot Welding
 - (ii) Seam Welding
 - (iii) Flash Welding
4. (a) What are the various types of automatic lathes in the multi-spindle category? Explain their differences and applications. Is the productivity of multi-spindle machines higher compared to single spindle automatic lathe? 7
- (b) Explain the various operations generally performed in a Sheet-metal shop. 3

J-1438

(Contd.)

- (c) Explain the advantages and disadvantages of compound dies over progressive dies. 4
5. (a) Explain Numerical Control (NC) machine tools. What are the classifications of NC machines? 7
- (b) How do you define a Machining Centre? How is cutter compensation given in the case of a machining centre? 7
6. Write short notes on any TWO of the following:
 - (i) Surface grinding
 - (ii) Taper turning methods
 - (iii) Capstan and Turret lathes
 - (iv) Laser Beam Welding

7X2=14

J-1438

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 23UMEC1158

BE II (ME)

3

MT

SECOND B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(III SEMESTER CBCS)
3ME42A (ME) : MATERIALS TECHNOLOGY

Time - Three Hours

Maximum Marks - 70

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) What type of lattice structure is usually indicative of ductility in metals? 3
- (b) A cube has six equivalent faces. Assuming a definite orientation of the cube, write down the Miller indices of the six facial planes of the cube. 3
- (c) A metal has a BCC crystal structure and an atomic radius of 1.732 Å. Calculate the number of atoms per square centimeter of (100) planes. 4

Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 23UMEC1158

BE II (ME)

3

PMT

SECOND B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(III SEMESTER CBCS)
3ME45A (ME) : PRODUCTION MACHINE
TOOLS

Time - Three Hours

Maximum Marks - 70

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) What type of work holding devices are generally used in a lathe? 4
- (b) Describe the operation of the quick return motion in a Mechanical Shaper. 5
- (c) Explain briefly the construction of a radial drilling machine with emphasis on how the requisite motions are obtained. 5

14.66

MBM UNIVERSITY JODHPUR
3ME21A(MA): MATHEMATICS
First Mid-term

Time: 1 Hour (Attempt any four)

MM: 10

1. The ordinates of the normal curve are given by the following table:

x:	0.0	0.2	0.4	0.6	0.8
y:	0.3989	0.3910	0.3683	0.3332	0.2897

Find: (By Stirling formula)

(a) $y(0.43)$ (2.5)

2. Use Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule to evaluate the following:

$$\int_0^1 \frac{dx}{1+x^2}$$

Hence obtain the approximate value of π in each case. (2.5)

3. Find the real root of the equation $x^3 - 3x - 5 = 0$ correct to four places of decimals by Newton Raphson method. (2.5)

4. Solve the following system by Gauss-Seidal iteration method:

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

(2.5)

5. Use Euler's modified method with one step to solve $\frac{dy}{dx} = x^2 + y$ with $y(0) = 0.94$. Find $y(0.1)$ to five significant figures $y(0) = 0.94$ (2.5)

6. Using Lagrange's Interpolation formula, find the value of y when $x = 10$ from the following data:

x:	5	6	9	11
y:	12	13	14	16

(2.5)

11-3-5
11-8

$P(x) = 10$
 $f(y) = 10$
 $y = 10$

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
First Mid-Term Examination Session 2023-24
ME-203A: Kinematics of Machines

Time: 1 hour

Maximum Marks – 20

Q. 1(a). An open belt running over two pulleys of diameters 200 mm and 600 mm connects two parallel shafts placed at a distance of 2.5 m. The smaller pulleys rotates at 300 rpm and transmits 5 kW. The coefficient of friction between the belt and the pulley is 0.3. Determine: (i) length of belt; (ii) initial tension; (iii) minimum width if the safe working tension is 12 N/mm width. (6)

~~(b) Explain the following: (i) idler pulleys; (ii) intermediate pulleys; (iii) loose and fast pulleys; (iv) guide pulleys.~~ (4)

Q. 2 (a) A multiple plate clutch transmits 55 kW of power at 1800 rpm. The coefficient of friction is 0.1. The inner radius is 80 mm and is 0.65 times the outer radius. If the intensity of pressure is not exceed 160 kN/m², determine the number of pair of friction surfaces needed to transmit the required torque. (6)

~~(b) What is friction circle? Derive an expression for its radius.~~ (4)

MBM UNIVERSITY
MECHANICAL ENGINEERING DEPARTMENT
Mid Term Assessment – 02 November, 2023

Subject Code: 3ME42A(ME)

Subject: Materials Technology

Max. Marks: 20

1. What is difference between amorphous and crystalline materials? What are the various types of crystalline structures. Draw neat diagram of each. (7)
2. Explain and formulate following with respect to unit cell. (7)
 - a) Number of atoms per unit cell.
 - b) Atomic radius
 - c) Atomic packing factor.
3. Explain various types of imperfection in crystals. Explain types of dislocations in crystal materials with suitable diagram. (6)

MID TERM II (B.E. III SEM 2023-24)
ME:204a Mechanics of Solids
DEPT. OF MECH. ENGG., MBM UNIVERSITY

TIME: 60 MIN MM:30

Q1. A column 150 mm x 200 mm is 6 m-long, both ends fixed $E = 1.75 \times 10^5 \text{ N/mm}^2$.

a. Find Crippling load b. Safe load, if $Fos = 3$

(6)

Q2. At a point within a body subjected to two mutually perpendicular stresses 60 N/mm^2 and 30 N/mm^2 tensile and 45 N/mm^2 of shear stress. Find σ_1 , σ_n and σ_r on a plane inclined at 30° to normal cross-section. (8)

Q3. Derive deflection equation

(6)

OR

Q3. A hollow shaft is to transmit 300 KW at 80 rpm. If the shear stress is not to exceed 60 N/mm^2 and $ID = 0.6 \text{ OD}$, find internal dia and external dia assuming minimum torque is 1.4 times the mean torque. Explain theory of pure bending and state the assumptions made. (6)

Q4. Write question no. and correct option in answer sheet. (10 $\rightarrow$ 2x5)

		a.	b.	c.	d.
Q1	Rankine formula is good for	Short column	Long column	Medium column	Both 'a' and 'b'
Q2	Maximum slope in cantilever beam with UDL 'w' is	$wL^3/9EI$	$wL^3/16EI$	$wL^3/3EI$	$wL^3/6EI$
Q3	A column with minimum equivalent length has	both ends hinged	one end fixed and the other end free	one end fixed and other end hinged	both ends fixed
Q4	The load at which a vertical compression member just buckles is known as	Critical load	Crippling load	Buckling load	All of these
Q5	Which of the following statements is true for a simply supported beam?	Deflection at supports is maximum.	Deflection is maximum where slope is maximum.	Slope is minimum at supports	Deflection is maximum where slope is zero.

2. A quick return mechanism is shown in fig. 1. Link 2 rotates uniformly at 20 rad/sec. Draw the velocity diagram and acceleration diagram and obtain the total acceleration of slider B.

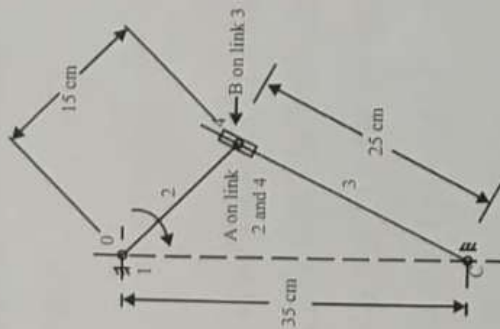


Fig. 1

3. (a) Explain the Pantograph mechanism with neat sketch. 7
- (b) For the Universal joint obtain the expression for ratio of speed of driving and driver shaft. 7
4. (a) What is the limiting angle of friction? Explain. 4
- (b) Calculate power lost in overcoming friction and number of collars required for the thrust bearing whose contract surfaces are 20 cm external radius and 15 cm. in internal radius. The coefficient of friction is 0.08. The total axial load is 30 kN. Intensity of pressure is not to exceed 3.5 bar. Speed of shaft is 420 rpm. 10

J-1435

2

5. (a) What are the advantages of rolling contact bearings over other bearings? 4

(b) A car engine has its rated output of 10 kw. Maximum torque developed is 100 N-m. The clutch used is of single plate type, having two active surfaces. Axial pressure is not to exceed 0.85 bar. External diameter of the friction plate is 1.25 times the internal diameter. Determine the dimensions of the friction plate and the axial force exerted by the springs. Assume $\mu = 0.3$. 10

6. (a) What is creep in belts? What is its effect in belt drive? 4

(b) A leather belting mass 1 gm/cm³ has a maximum permissible tension of 21 bar. Determine the maximum power that can be transmitted by a belt 25 cm x 1.1 cm, if the ratio of friction tension is 2. 10

7. (a) Sketch a Bush and Roller Chain. Give the classification of chains. 4

(b) 100 kw is transmitted by rope drive through a 160 cm diameter 45° grooved pulley running at 200 rpm. Angle of overlap is 140° and coefficient of friction between the pulley and the rope is 0.25. Mass of rope is 0.7 kg/m. and it can withstand a tension of 800 N. Considering the centrifugal tension induced, determine

(i) the number of ropes needed for transmitting the required power.

(ii) the tension in the rope before starting. 10

J-1435

3

6. (a) A steel bar of rectangular section $30 \text{ mm} \times 40 \text{ mm}$ one end fixed and other is free at each end is subjected to axial compression. The bar is 1.75 m long. Determine the buckling load and the corresponding axial stress using Euler's formula.

Determine the minimum length for which Euler's equation may be used to determine the buckling load, if the proportional limit of the material is 200 N/mm^2 .

Take $E = 2 \times 10^5 \text{ N/mm}^2$.

14

7. Two planes AB and BC which are at right angles carry shear stresses of intensity 35 N/mm^2 while these planes also carry a tensile stress of 140 N/mm^2 and a compressive stress of 70 N/mm^2 respectively. Determine the principal planes and the principal stresses. Also determine the maximum shear stress and the planes on which it acts.

14

2. Explain the difference between the following pair of terms:-

- (a) Elastic limit and proportional limit 3
(b) Toughness and resilience 4
(c) Elasticity and plasticity 3

3. Define: Ferrite, Cementite, Austenite, Pearlite, Martensite and Bainite. What are their general characteristics? How are they obtained? 7

(b) What is meant from 'Critical Temperature Range'? What practical significance does it have? 3

4. (a) What is the major difference in the purpose of annealing and normalizing? 4

(b) What is the importance of hardenability of steel as applied to heat treatment? 3

(c) How do the composition and structure of a high strength constructional steel differ from that of a tool steel? 3

5. (a) High-carbon-high-chromium steels are widely used in industry. What are their important characteristics and applications? 4

(b) How would you control the composition of cast iron produced by Cupola? 3

J-1434

2

(Contd.)

(c) What are the advantages of aluminium alloys? Where are they used? 3

6. (a) Comment upon the metallurgical structure of bearing alloys and compare it with that of metals and homogeneous alloys. 3

(b) What are Refractory Materials? Give a few examples of refractory materials. 4

(c) What are the different types of composites available? Give typical examples of each group. 3

7. (a) What are Compaction and Sintering? Explain with suitable examples. 4

(b) What are Smart Materials? Give their applications. 3

(c) Differentiate between Nitriding and Cyaniding. 3

J-1434

3

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Roll No.23VMECH158

BE II (ME)

3

KOM

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME43A (ME) : KINEMATICS OF MACHINES

Time - Three Hours

Maximum Marks - 70

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Differentiate between Mechanism and Machine. 4

(b) What do you mean by Inversion of mechanism?

5 Explain any two inversions of double slider crank chain mechanism. 10

J-1435

1

(Contd.)

BE II YEAR III SEM. MID TERM-I
3ME45A(ME): PRODUCTION MACHINE TOOLS

TIME: 01 HOUR

MM: 20

Q1. Explain Constructional details and operations on

(a) Centre lathe

(b) Universal milling machine

(2x5 = 10)

Q2. Differentiate between Jigs and Fixture. Write the principles of location.

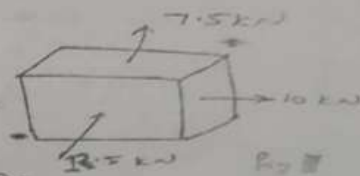
(5)

Q3. Explain MIG welding with sketch. Differentiate between SMAW and MIG welding. (5)

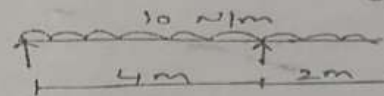
MID TERM I (B.E. III SEM 2023-24)
ME:204a Mechanics of Solids
DEPT. OF MECH. ENGG., MBM UNIVERSITY

TIME: 60 MIN MM:30

Q. Find the change in volume of the cube of 100X100 mm as shown in fig if $1/m=0.3$ and $E_{\text{Steel}} = 2.1 \times 10^5 \text{ N/mm}^2$. (6)



Q. Draw the SFD and BMD for the beam shown: (8)



Q. Explain theory of pure bending and state the assumptions made. (6)

Q. Write question no. and correct option in answer sheet. (10 = 2x5)

	a.	b.	c.	d.
If internal resistive force is 'F' and 'A' is resisting area, stress is defined as	F/A	F	A/F	AxF
Figure out odd term	Young's modulus	Compressive stress	Shear stress	Tensile stress
If change in volume is ΔV and original volume is V, Volumetric Strain is defined as	$\Delta V/V$	$\Delta V + V$	$V - \Delta V$	$\Delta V \times V$
The total extension of a taper rod of length 'L' and tapering diameters ' D_1 ' and ' D_2 ', subjected to a load (P), is	$4PL/\pi D_1 D_2$	$PL/\pi D_1 D_2$	$8PL/\pi D_1 D_2$	$PL/4\pi D_1 D_2$
Extension of a bar of uniform cross section 'A' of length 'L' due to self weight 'W' is given by	$WL/2AE$	$WL/4AE$	$WL/8AE$	WL/AE

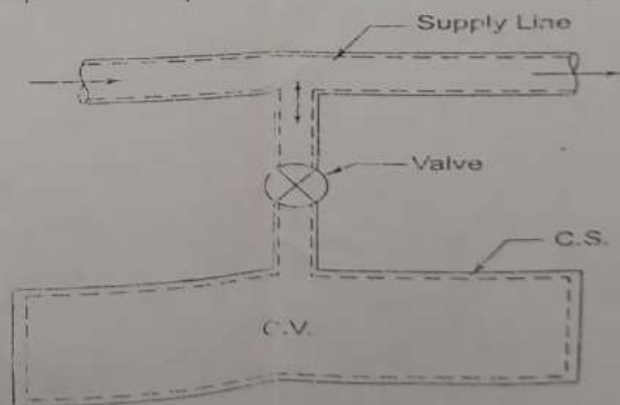
1st Midterm
ME 201A: Engineering Thermodynamics

Date 02/11/2023

Exam Duration: One Hour (12 PM to 1 PM)

Maximum Marks: 20

1. State Kelvin-Planck and Clausius' statement. [2+2]
2. State entropy principle. [1]
3. Derive the expression of exergy balance for a closed system. [6]
4. What is stoichiometric air fuel ratio. What is excess air. Write combustion equation for fuel $C_{10}H_{22}$. [3]
5. A rigid, insulated tank/bottle is initially evacuated. A valve connected with the supply line is opened and the tank is charged with air (ideal gas) until the final pressure inside the tank reaches the pressure in the supply line. Derive the expression of final temperature inside the tank that- [Final temperature = γ * temperature of the tank] [6]



2nd Mid Term

3ME42(42): Material Technology

Date: 07/12/2023

Exam Duration: 1 Hour (02:30 pm to 03:30 pm)

M.M.: 20

Question 1.

- (A) — Explain recovery, recrystallisation and grain growth in metals. (5)
(B) — What is meant by hot and cold deformation in metals? (5)

Question 2.

- (A) — What is the importance of heat treatment for metals. (4)
(B) — Explain term annealing, tempering, normalizing and hardening. (6)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 3ME21A (ME)

BE II (ME)

23UMEC1158

3

Math.

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME21A (MA) : MATHEMATICS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Solve

$$\frac{dx}{dt} - x + 2y = 0; \frac{dy}{dt} - 5x - 3y = 0$$

7

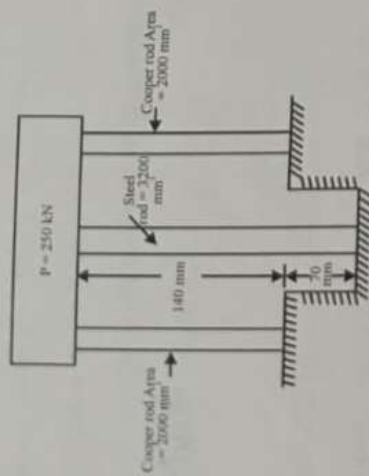


Fig. 1

10

(b) A steel bar of 30 mm diameter is loaded as shown in

Fig. 2. Determine the stresses in each part and the total elongation. Take $E = 210 \text{ GPa}$.

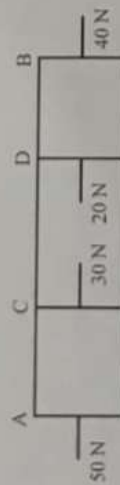


Fig. 2

4

2. Draw SFD and BMD for the beam shown in Fig. 3

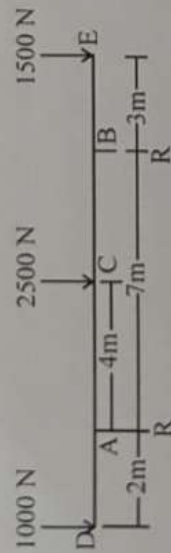


Fig. 3

14

3 (a) A timber joist of 200 mm width and 400 mm depth is used as a cantilever beam of 3 meters length and loaded

J-1436

2

with a concentrated load at its free end in addition to its self-weight, so that the bending stress does not exceed 7 N/mm^2 . Find out the maximum value of the applied load. Unit weight of timber is 5 kN/m^3 .

(b) What do you mean by Section Modulus? Find an expression for section modulus for rectangular, circular and hollow circular section.

4

4. For the loaded beam shown in Fig. 4 find (i) the slopes at A, C and D (ii) the deflection at C and D and (iii) the maximum deflection. Take $E = 200 \text{ kN/mm}^2$ and $I = 9.25 \times 10^7 \text{ mm}^4$.

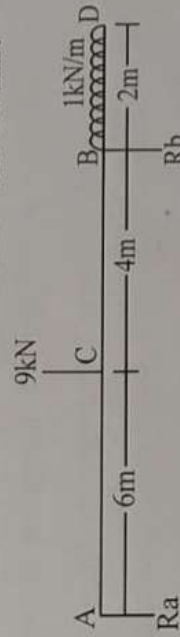


Fig. 4

14

5. (a) Explain significance of Rankine formula for Columns.

(b) A hollow shaft is to have an outside diameter d and inside diameter $\frac{d}{2}$. Calculate the minimum value of d if it is to transmit 375 kW at 100 rpm with a working stress of 40 N/mm^2 . Determine the twist in a length equal to 10 times the external diameter. Take $C = 8 \times 10^4 \text{ N/mm}^2$.

(Contd.)

3

J-1436

Quiz (M.M. 5)

Each question carries one mark.

1. When a gas is throttled, the maximum temperature drop or cooling will occur if the initial state lies on the
 - (a) Saturation curve
 - (b) inversion curve
 - (c) constant enthalpy curve
 - (d) heating region
2. The vapour pressure of a liquid at any arbitrary temperature can be estimated approximately with the help of
 - (a) Gibbs equation
 - (b) Joule-Kelvin equation
 - (c) Clausius-Clapeyron equation
 - (d) Maxwell equation
3. Cylinder clearance in a compressor should be
 - (a) as large as possible
 - (b) as small as possible
 - (c) about 50% of swept volume
 - (d) about 100% of swept volume
4. The specific gas of an ideal gas depends on its
 - (a) Density
 - (b) Molecular weight
 - (c) Pressure
 - (d) Temperature
5. Volumetric efficiency is
 - (a) the ratio of stroke volume to clearance volume
 - (b) the ratio of the air actually delivered to the amount of piston displacement
 - (c) reciprocal of compression ratio
 - (d) index of compressor performance

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
First Mid-Term Examination Session 2023-24
ME-203A: Kinematics of Machines

Time: 1 hour

Maximum Marks – 20

Q. 1(a). An open belt running over two pulleys of diameters 200 mm and 600 mm connects two parallel shafts placed at a distance of 2.5 m. The smaller pulleys rotates at 300 rpm and transmits 5 kW. The coefficient of friction between the belt and the pulley is 0.3. Determine: (i) length of belt; (ii) initial tension; (iii) minimum width if the safe working tension is 12 N/mm width

$\propto bT$

(6)

(b) Explain the following: (i) idler pulleys; (ii) intermediate pulleys; (iii) loose and fast pulleys; (iv) guide pulleys.

(4)

Q. 2 (a) A multiple plate clutch transmits 55 kW of power at 1800 rpm. The coefficient of friction is 0.1. The inner radius is 80 mm and is 0.65 times the outer radius. If the intensity of pressure is not exceed 160 kN/m², determine the number of pair of friction surfaces needed to transmit the required torque.

(6)

(b) What is friction circle? Derive an expression for its radius.

(4)

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
Second Mid-Term Examination Session 2023-24
ME-203A: Kinematics of Machines

Time: 1 hour

Maximum Marks – 20

Q.1(a). In a four bar mechanism ABCD, the link AD is fixed link of 120 mm long, crank AB is 30 mm long and rotate uniformly at 100 rpm clockwise, while the link CD is 60 mm long and oscillates about fixed point D. The link BC is equal to length AD. Find the angular velocity of link DC when angle BAD is 60° .

(b) What do you mean by instantaneous centre? Explain the properties of instantaneous centre.

Q.2 What is the difference between lower and higher pairs? Give examples of each type.

(b) Sketch and explain the various inversion of single-slider crank chain mechanism.

(c) What do you understand by terms machine and mechanism? State how these differ from each other.

3ME21A (MA): MATHEMATICS
SECOND MID TERM

Time: 1 hour

(Attempt any four)

MM:19

Q.1. Find Laplace $\sin \sqrt{t}$. Also find Laplace $\frac{\cos \sqrt{t}}{\sqrt{t}}$. (2.5)

Q.2. Find $L^{-1} \left[\frac{s^2}{(s^2 + a^2)^2} \right]$ by convolution theorem. (2.5)

Q.3. Solve $(D^2 + 1)y = t \cos 2t$, $y(0) = 0 = y'(0)$, $t > 0$ by Laplace transformation (2.5)

Q.4. Find the Fourier Transform of (2.5)

$$f(x) \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$$

Q.5. Find the Fourier sine transform of $\frac{1}{x}$. (2.5)

Q.6. Solve $\frac{dy}{dx} - 7x + y = 0$, $\frac{dy}{dx} - 2x - 5y = 0$. (2.5)